

Course Syllabus

Course Title: EM 509 - ENGINEERING PROJECT MANAGEMENT

EMU COLLEGE OF TECHNOLOGY

School of Engineering Technology

Prerequisites: None

Credit hours: 3 Semester Hours

Instructor: Dr. Muhammad S Ahmed

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Office Hours: *E-mail me to schedule phone appointment.*

Wednesday 9:00AM – 3:00PM

Course Description

A study of the planning and management activities associated with the implementation of a manufacturing or engineering project.

Topics include: identification and sequencing of events, time and cost estimating, resource utilization, and monitoring/evaluation of plans.

Required Texts and Software

Required: Gray, C., & Larson, E. (2011). Project management: The managerial process w/ Student CD and MS Project CD (5th ed.). New York: McGraw-Hill Higher Education. (ISBN-10: 0-07-340334-2; ISBN-13: 978-0-07-340334-2)

Recommended: Chatfield, G., & Johnson, T. (2010). Microsoft Project 2013 Step by Step. Redmond, WA: Microsoft Press. ISBN: 978-0735669116

Software: MS Project 2013: Install MS Project 2013 as an EMU student via the following website: <https://app.emich.edu/dreamspark/> - you will need to log in the site with your EMU ID and password. This is not a permanent installation but will be available throughout this semester.

You are welcome to use an earlier version of the software that you are familiar with; however, the examples will be offered using MS Project 2013.

Erik Larson's MS Project 2010 Tutorials can be found in the Doc Sharing site.

Microsoft - Internet Explorer. Issues were reported related to watching some of the videos included in the course content with Mozilla Firefox.

Course Objectives

At the completion of this course, the student should be able to:

- A. Identify, describe, and apply major concepts and methods of engineering project management.
 - B. Analyze project objectives, determine tasks, allocate resources, and construct schedules.
 - C. Describe ongoing project management techniques.
 - D. Utilize project management software.
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Course Outline and Schedule

PROJECTED SCHEDULE (Subject To Change)

See - Due Dates - File Under - Course Home - Tab for Unit Beginning and End Dates

Unit requirements are due to be submitted by 11:59 p.m. on the Unit End Date

Unit 1

Introduction.

Modern project management.

Organization strategy and project selection.

Unit 2

Organization: structure and culture.

Defining the project.

Unit 3

Estimating project times and costs.

Developing a project plan.

Unit 4

Managing risk.

Scheduling resources and costs.

Unit 5

Reducing project duration.

Leadership: Being an effective project manager.

Managing project teams.

Unit 6

Outsourcing: Managing inter-organizational relations.

Progress and performance measurement and evaluation

Unit 7

Project closure.

International projects.

Oversight.

Unit 8

An Introduction to Agile Project Management

Project Management Career Paths

Final exam.

General Policies

- A. Students are encouraged to discuss problems related to the class with the instructor via telephone or email.
- B. Academic Dishonesty: The policy of the School of Engineering Technology is that any student found to have engaged in any activity constituting academic dishonesty will receive an "E" for the course in which the activity occurred. This policy relates to all forms of work associated with the course requirements (tests, lab work, papers and other assignments). Academic dishonesty includes: cheating on quizzes and exams, copying, or taking credit for another's work, plagiarism, etc. Graduate students enrolled at EMU should be aware of academic dishonesty policies and consequences, and examples of what constitutes academic dishonesty, from EMU Academic Affairs, EMU Student Judicial Services, EMU Graduate School, and the School of Engineering Technology.

Our managerial approach regarding minor occurrences of academic dishonesty is to inform the involved student(s) about the problem, give a warning to the student(s) of the possible consequences, and share an example of the problem with the rest of the class, as a learning experience for the whole class. The consequences may be more severe (such as failure of the assignment or the entire course), and without a warning, for second and/or major occurrences of academic dishonesty.

Turnitin.com is a web-based service that compares submitted student text against a database of millions of previously published documents, including those on the public Internet, a proprietary collection of published articles, as well as every student paper ever submitted to Turnitin.com, including previously submitted EMU papers. When similarities between student text and an existing document are found, Turnitin highlights those similarities in an annotated document, displaying both the student text and the original source(s). Detection of similarities does NOT mean plagiarism has occurred as the overlapping text might be properly referenced and cited. In this course I may submit your assignments or you may be asked to submit assignments to Turnitin.com for textual similarity review.

All Chapter Summaries must be submitted via Turnitin.com.

- C. Writing Style and Format: It is the policy of the School of Engineering Technology (SET) that students should follow the APA (American Psychological Association) technical writing style and format guidelines for papers submitted in all IT classes. Some professors will emphasize use of APA guidelines more than others. It is recommended that you get a copy of the 6th edition of the APA Publication Manual as soon as possible, as you will use it throughout your degree. You should especially be familiar with traditional and electronic referencing guidelines for quoting, paraphrasing, and referencing information that you use, to help avoid Academic Dishonesty problems, as mentioned above. See <http://www.apastyle.org/styletips.html> and <http://www.apastyle.org/eleceref.html> for more specific information and examples.
- D. Student and Exchange Visitor Information Service (SEVIS): For students who hold an F or J visa, to avoid being out-of-status, it is your responsibility to report within 10 days to the Office of International Students (OIS), King Hall, any change in name, current address of residence, academic status (full or part-time enrollment), date of program completion, program/major change, level change (undergraduate to graduate), change in funding source (employment or graduate assistant position), probation or disciplinary action leading to suspension due to conviction of a crime. Effective January 1, 2003, EMU has 21 days from the date of the reported event to inform the U. S. Department of Homeland Security, Bureau of Border and Transportation Security and the Bureau of Citizenship and Immigration Services - SEVIS - database (functions of the formerly known as Immigration and Naturalization Service are a part of two new offices effective March 1, 2003).

OIS enters the data into a real-time database and new documents or I-20s are then issued to you. Remember that dropping or withdrawing from course(s) may put you below full-time enrollment. Before a drop/withdrawal change is made that reduces enrollment below full-time status, you must receive permission from the OIS. If you plan to transfer to another university, this process must begin before the end of your last semester at EMU, contact the Office of International Students. Noncompliance could result in actions leading up to a fine, arrest, or deportation per the U.S. Government.

- E. Incomplete (I) will be given only in accordance with EMU's guidelines please follow this link for details - http://www.emich.edu/registrar/registration_info/grade_policy.php
- F. Quizzes and Exams: Most of the quizzes and exams in this course are timed, based on an easy standard of 120 seconds (two minutes) per question. The typical standard for professional-level exams, such as certification or licensure exams, is about 60 to 90 seconds per question. What makes it even easier for you, is that your quizzes and exams in our course are very focused on specific information from sections of specific books--whereas certification and licensure exams cover a very wide body of professional knowledge, covered by numerous books and other material.

Like some professional exams, your quizzes and exams in this class are open-book (of course)--and, like those professional exams (and regular classroom exams), you should prepare for taking your quizzes and exams in advance, by reading the material to be covered, taking notes, and applying other test-preparation studying techniques.

- G. Incomplete Grades: An "I" grade may be awarded when illness or other unavoidable extenuating circumstances prevent completion of a course, provided that academic performance for the completed portion of the course merits a grade of B or better.

Participation Guidelines

Here are some guidelines for making things go smoothly.

- A. Access to Course: Our course consists of 8 units, which are scheduled from Mondays through Sundays. Most units will be scheduled across two weeks, but some units may be slightly shorter for various reasons. Assignment due dates for each unit are located in the "Assignment" tab of the unit. As a general rule of thumb most assignments are due on a Sunday night, by 11:59 PM ET. We will usually open access to each unit early, on Fridays, so that you can get an early start working on it if you like; therefore, you will have three weekends available for working on most of the units. All completed units will remain open for review, until the end of our class.
- B. Participation: In online courses, students seek interaction with other students and with the instructor. Since there is no face-to-face meeting, this interaction becomes very important to effective learning online. As such, good class participation is essential and will account for a significant part of your grade. [Plan to contribute to every discussion topic with a "quality" contribution -- i.e. a response that expresses clear thinking, and that is relevant to the discussion. This usually involves more than just "I agree."]
- C. Online Conduct: I believe that discussion of the following is probably not very necessary. Nevertheless, for the sake of clarity, please realize that the University and I cannot and will not be responsible for the content of any personal messages that are sent from one student directly to another student using the online email delivery system. It is expected that all students will adhere to accepted codes of ethical, personal, and civil conduct when conversing online, using email, or engaging in any real time online chat sessions. Failure to abide by such codes of conduct and etiquette may result in withdrawal from the course and a failing grade.
- D. Email to Classmates: Students are free to use the "email the class" feature on the course web site. Try to keep the messages sent to the whole class related to course content. Also, when emailing the class, generally it is best NOT to email phone numbers or addresses unless it is being emailed directly to a specific individual or the instructor. This helps maintain privacy.
- E. Email to Instructor: For course-related communications, please send an email using the address listed under the course home tab in the course syllabus. Also, in the subject line of your email messages, include the: initials of your full name, course number, and the subject of the message (such as the name of an assignment that is attached to your email); for example, Subject: BWB509Homework1.
- I can't be on call for your class 24 hours a day, but will be checking in as often as possible. I expect to check EM 509-related email before 12 noon on several days during the week. Due the amount of email I will receive from students, as well as other work and travel commitments, I may not be able to respond to emails more than once or twice a week--although I will try to do so more frequently. So, plan accordingly. If you wait until the last minute to do an assignment, and then email a request for assistance with a problem, neither I nor a fellow student may be able to respond quickly enough to help you.
- F. Submittals and Attachments: All assignments are to be turned in via Dropbox feature of our course. Go to Dropbox by clicking the tab on the top of your screen, and then click Submit an

Assignment link; choose the corresponding basket to submit your assignments. Once an assignment is submitted, it will appear in your Outbox until I grade it and return it to your Inbox with points you got. Typically I return submitted assignments to your Inbox in two days or three after a unit has been complete.

There are so many different word processing programs that assignment attachments can become a problem in online courses if there is no standard policy set for them. It is preferred that you send any paper as an attachment in MS Word. Formats that are compatible with MS Word include earlier Word versions; Microsoft Works, Word Perfect 5 or 6, and Rich Text Format are also acceptable. If necessary, you can compose the work in your non-compatible word processor and simply cut and paste it into the text message--although this may cause a problem if formatting is important. Submit MS Project software assignments in MS Project format. You may also send single or multiple files in .zip compressed format.

- G. Grammar, Spelling, Writing Style, and Format: We all tend to become somewhat lax when posting responses to threaded discussions or when sending email messages. Grammar and spelling is important to me. I am not a fanatic about it, but I would appreciate your composing threaded discussion responses or email messages in a word processing program where you can check spelling, edit, and revise your writing. Then you can cut and paste the work into the threaded discussion or email message. You can do the same for other coursework and thereby help with NOT tying up Internet Service Provider lines.

Good grammar, spelling, writing style, and APA format is required for formal papers, per SET guidelines.

- H. Assignment Delivery and Technical Problems: This is a Project Management course. Project Managers are expected to facilitate deliverables and hit deadlines. This is also a Masters level course in Engineering Management. The EM program is designed to create people who are able to manage others. One first has to be able to demonstrate that they can manage themselves. This class places an emphasis on handing assignments in on time. Assignments must be submitted before midnight (11:59 PM ET US) on the due date displayed in the "Assignment" tab. Assignments will NOT be accepted late. Since you will have more than two weeks and three weekends available for completing your work throughout most of the course, I expect you to get work done on time. Practice Risk Management of unforeseen risk. Start your homework early and work on it day by day so that you are not waiting till the weekend that it is due to finish your assignment, and then encounter a problem causing you to miss your deadline.

We encourage you to make hard copies of your assignments and communications with your instructors. Computer crashes and disasters are bound to happen. Keep in mind that we do not view ISP problems as an excuse for turning work in late. At the first sign of problems with email, contact your ISP immediately. If your problems are not cleared up within 12 hours, send in the assignment in one of the following ways:

1. Overnight express hard copy and on disk.
2. Make arrangements over the phone with me.

Note: It is your responsibility to access the course on another computer in the event of hardware or software problems. This can be done with a friend's computer, or one at Kinko's or the Public Library. Our guidance from the university is to not consider any excuse for missing assignments due to technical difficulties as legitimate. Students should work on their

assignments early, that way if there are technical problems the student will be able to address the issues and hand in the assignment in on time.

- I. Personal Involvement and Time: Students often have the misperception that taking a course online is easier since they do not have to physically go to a classroom. When taking an online course there are some things that become easier for students. For example, it becomes easier NOT to go to class. It also becomes easier to get behind. This medium gives students flexibility in attending class--but remember, while taking a course online may be more convenient, it is NOT necessarily easier!

Be aware of the general guidelines to instructors for student workloads: 45 hours of work per credit hour per semester. This is a 3 credit class, so I expect, on average, that students will spend 135 hours of total work time in this course. Students with more graduate-level and online learning experience tend to be more efficient and require less time--some students will require a great deal more time to get their school work completed. Be prepared for the work involved, as it may require 15 or more hours of your time each week.

Remember, this is a course for the Engineering Management program. Working well together in teams, and making sure that you get your assignments done for your team are essential for project success. Engineering Managers are responsible for planning, organizing, staffing, leading, and controlling engineering projects and personnel. They must lead by example, and make sure that objectives are met and completed on time. Many of our EM students have very busy professional lives and have significant work and family responsibilities in addition to going to school--yet they do their school work well, and get it done on time (often early). If a student in our program cannot organize themselves well enough to get their school work done on time, then they will certainly have a problem organizing and leading other engineers and engineering projects, and probably should not be in this program. As such, I will give little leeway when it comes to assignments that are not submitted on time--and will suggest to chronically late students that they seek a different degree and profession.

- J. EMU Writing Support: The University Writing Center (115 Halle Library; 487-0694) offers one-to-one writing consulting for both undergraduate and graduate students. Students can make appointments or drop in between the hours of 10 a.m. and 6 p.m. Mondays through Thursdays and from 11 a.m. to 4 p.m. on Fridays. The UWC opens for the Winter 2015 semester Monday, January 11, 2016 at 10 a.m. Details can be found here: <http://www.emich.edu/uwc/>

Information about the UWC's satellite locations and timing can be found here:

<http://www.emich.edu/uwc/uwc-online/index.php>. Information about the UWC's Online support can be found here: <http://www.emich.edu/uwc/uwc-online/index.php>. Students seeking writing support at any location of the University Writing Center should bring with them a draft of what they are working on and their assignment sheet.

The Academic Projects Center (116 Halle Library) also offers one-to-one writing consulting for students, in addition to consulting on research and technology-related issues. The APC is open 11 a.m. to 5 p.m. Mondays through Thursdays for drop-in consultations. Additional information about the APC can be found at <http://www.emich.edu/apc>.

Grading Policies

- Exams – 40%
 - Unit Exams – 20% (2.86% per unit)
 - Final Exams – 20%
- Projects – 35%
 - Computer Exercise (The POM+ Project) – 17.5%
 - Personal Project – 17.5%
- Threaded Discussion – 10%
- Chapter Summary – Outlines – 15%
- Extra Credit Potential - 4%

Evaluation

Course grades will be determined by the percentage of total points attained for various work assignments completed throughout the semester, based upon the following scale:

100 to 93% = A

92.99 to 90% = A-

89.99 to 87% = B+

86.99 to 83% = B

82.99 to 80% = B-

79.99 to 77% = C+

76.99 to 73% = C

72.99 to 70% = C-

No D grades: based on Grad School requirements

69.99 and below = F

Final course grades will be based primarily (but not necessarily only) on the quantitative scale shown above--but are subject to adjustment based on the judgment of your professor in consideration of qualitative factors, including academic dishonesty and displays of other unethical, unprofessional, or uncivil behaviors that are unbecoming of a good engineering manager and graduate student. (You must be prepared for the reality of professional life in business and industry--there is usually more to getting a good performance evaluation from your boss and peers than just production of work, other factors include: (a) personal professional skills, (b) professional communication skills, and (c) ethics and emotional intelligence.) (a), (b), and (c) above are of vital importance to becoming successful as an Engineering Manager, and why they are three of our EM Program Student Learning Objectives. We expect to see them demonstrated by students in our classes.

Activities

The general format of this class will be through online guidance from the instructor, threaded discussion, and submission of assignments, quizzes, a final project, and a final examination.

ONLINE GUIDANCE FROM INSTRUCTOR: Students will follow the directions given by the instructor via the e-college system and submit completed assignments electronically to the instructor.

THREADED DISCUSSION: Students will receive points for participation in a threaded discussion on a topic posed by the instructor that is related to that unit.

DEBATE CASE: **In some classes**, during your Engineering Management program, you will participate in Case Studies solutions and debating various issues. Read the Unit Debate-Case to learn how to analyze case study and participate in debate.

CHAPTER ASSIGNMENTS: Students will complete and submit reading outlines, answers to review problems, and answers to quizzes for each chapter assigned throughout the course.

MS PROJECT SOFTWARE ASSIGNMENTS: Students will complete preparatory assignments and a final project using the MS Project software included with the textbook.

FINAL EXAMINATION: **In most classes**, you will have an objective, online, open book, timed final exam.